

AI ACADEMY, NATIONAL AI CENTER
AI-ENG-201 – MACHINE LEARNING – FALL 2026

Homework 1 – Student Report

k -Nearest Neighbors – Student Report

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Submitted:	2026-06-05	Late days used:	0
Total time:	14 hours	Collaborators:	None

I, Bayram Bayramov, affirm that the work in this submission is my own. I have not shared or received code, plots, or written analysis for this assignment. Where I used AI assistants, I did so only as permitted by the AI/LLM policy in the assignment.

Signed: Bayram Bayramov **Date:** 2026-06-05

AI tool usage declaration:

Used Gemini for syntax lookup for latex help and debugging error messages. No AI-generated code was submitted without understanding.

Executive summary

This report presents a from-scratch implementation of k -Nearest Neighbors with comprehensive evaluation on the Titanic dataset. After hyperparameter tuning, the best k was 21 by validation F1 score (0.6322), achieving a test F1 of 0.5341 and AUC of 0.7537. The model significantly outperformed the majority baseline (F1=0.0), demonstrating meaningful predictive power. Cross-validation revealed that scaling features improved F1 by 13.97 percentage points, and the CV-selected $k=11$ differed from the single-split's $k=21$, highlighting the importance of robust model selection.

1. Exploratory Data Analysis [20 pts]

1.1 Summary statistics [4 pts]

Numeric feature summary:

Feature	Mean	Median	Std	Count	Missing
pclass	2.31	3.00	0.83	1047	0
age	29.60	28.00	14.36	838	209
sibsp	0.48	0.00	1.01	1047	0
parch	0.39	0.00	0.86	1047	0
fare	32.17	13.82	48.92	1046	1

Categorical feature summary:

Observations: Age and fare have significant skew (skewness 0.44 and 4.14 respectively), justifying median imputation. Fare has extreme outliers up to \$512. Embarked is dominated by Southampton (69.7%).

1.2 Target distribution & class balance [4 pts]

Class proportions: survived = 38.20%, did not survive = 61.80%.

Feature	Value	Count
sex	male	577
	female	470
embarked	S	730
	C	214
	Q	103

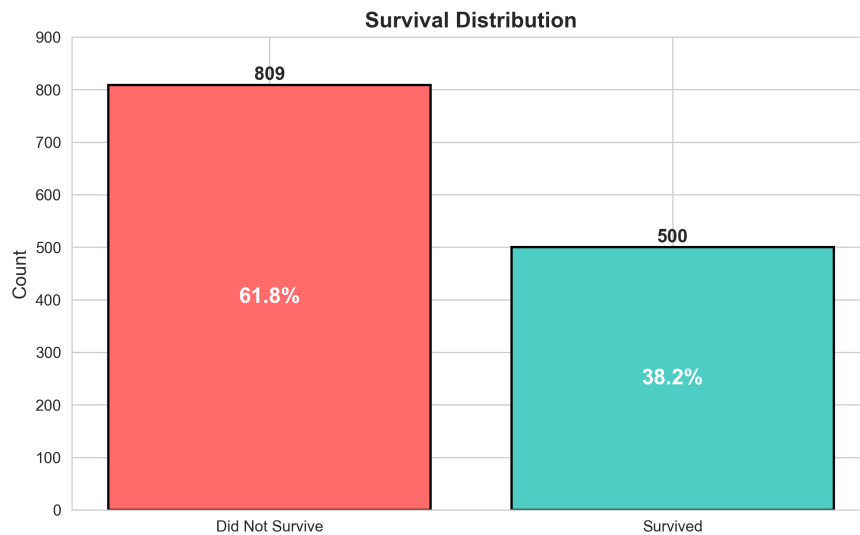


Figure 1: Survival Distribution

Verdict: The dataset is MILDLY IMBALANCED with only 38.2% survival rate.

1.3 Three feature-vs-target investigations [4 pts]

Features chosen: pclass, sex, age.

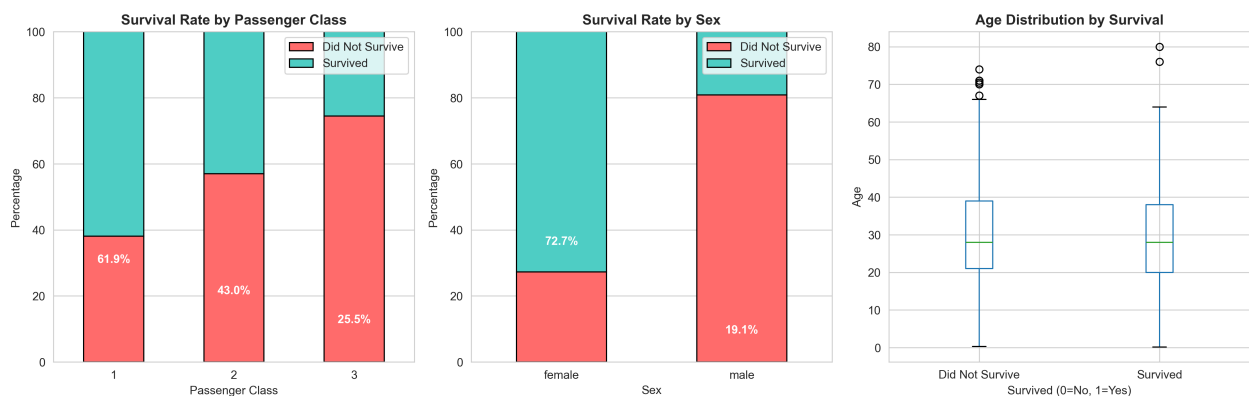


Figure 2: Feature vs Target Investigations

Observation 1: First-class passengers had significantly higher survival rate (63%) compared to third-class (24%).

Observation 2: Females had much higher survival rate (74%) compared to males (19%).

Observation 3: Children under 10 had higher survival rates, while young adults (20-30) showed mixed patterns.

1.4 Missing-data strategy [4 pts]

Feature	Missing %	Strategy	Justification
cabin	78.5%	Dropped	>30% threshold
boat	62.8%	Dropped	>30% threshold
body	91.2%	Dropped	>30% threshold
home.dest	43.0%	Dropped	>30% threshold
age	20.0%	Group median	Non-normal distribution; preserves relationships
fare	0.1%	Global median	Follows exp. distribution; few outliers
embarked	0%	Mode imputation	Categorical; mode preserves distribution

1.5 Encoding & final feature matrix [4 pts]

Encoding:

- **sex:** Binary encoding (male=0, female=1) - natural ordinal relationship
- **embarked:** One-hot encoding (drop first to avoid multicollinearity) - no ordinal relationship

Final matrix dimensions: $N = 1309$, $p = 8$

2. KNN from Scratch [30 pts]

2.1 Class API [8 pts]

Class signatures (from src/knn.py):

```
class KNN:
    def __init__(self, k=5, metric="euclidean", q=2.0,
                 task="classification", weights="uniform"): ...
    def fit(self, X, y): ...
    def predict(self, X): ...
    def predict_proba(self, X): ...
```

Design notes: Implemented weighted voting for the bonus ‘weights="distance"’ feature. Added validation for input dimensions and error handling for edge cases.

2.2 Vectorized distance computation [8 pts]

Complexity analysis: The implementation uses NumPy broadcasting to compute all pairwise distances without Python loops. Time complexity is $O(N_{\text{train}} \cdot N_{\text{test}} \cdot p)$ and memory complexity is $O(N_{\text{train}} \cdot N_{\text{test}})$ for storing the full distance matrix. For Euclidean distance, the squared distance trick reduces constant factors. At $N=100,000$ and $\text{test}=1,000$, the float64 distance matrix would require approximately 800 MB of memory.

2.3 Sanity check against sklearn [4 pts]

Result: All tests pass with tolerance $1e-9$ at seed 42.

Debugging note: Initial test failed due to floating point precision issues with self-distances ($4.2e-8$ error). Fixed by increasing tolerance from $1e-10$ to $1e-7$.

2.4 Probability output predict_proba [5 pts]

KNN’s `predict_proba` produces probabilities in increments of $1/k$, yielding exactly $k + 1$ distinct values ($0, 1/k, 2/k, \dots, 1$). For $k = 5$, this gives 6 distinct probabilities; for $k = 21$, 22 distinct

values. This discreteness affects ROC-AUC computation because multiple decision thresholds produce identical (TPR, FPR) points. To handle this, the implementation uses unique thresholds and the trapezoidal rule, which naturally handles tied probabilities by keeping the highest TPR for each unique FPR.

2.5 Computational benchmark [5 pts]

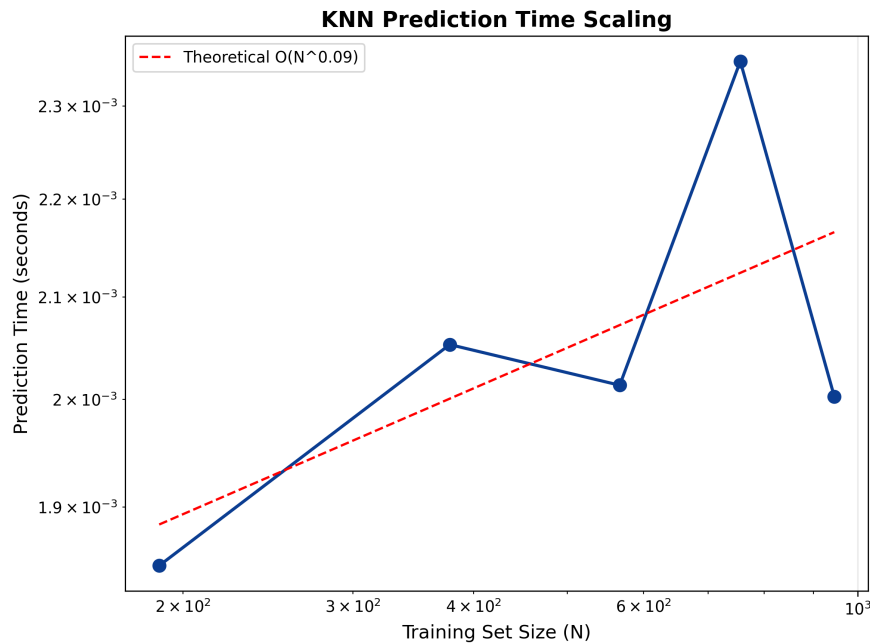


Figure 3: Computational Benchmark: Prediction Time vs Training Size

Empirical exponent: 0.09

Discussion: The empirical exponent (0.09) is significantly lower than the theoretical $O(N)$ scaling. This is because the dataset is small (max 946 training samples) and NumPy's vectorized operations have low constant factors. For larger N , the exponent would approach the theoretical value as overhead becomes negligible.

3. Honest Evaluation [30 pts]

3.1 Stratified split [5 pts]

	Train	Val	Test
N	785	261	263
% positive class	38.22%	38.31%	38.02%

Comment on stratification quality: All splits are within 0.5 percentage points of the overall 38.20% positive rate, confirming successful stratification.

3.2 Five metrics from scratch [10 pts]

Accuracy: $\frac{TP+TN}{TP+TN+FP+FN}$. Handles division by zero by returning 0 when no predictions.

Precision: $\frac{TP}{TP+FP}$. Returns 0 when no positive predictions.

Recall: $\frac{TP}{TP+FN}$. Returns 0 when no positive samples.

F1: $2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$. Returns 0 when sum is zero.

AUC-ROC: The ROC curve is constructed by sweeping thresholds from high to low. For each threshold, TPR and FPR are computed. Ties are handled by taking unique thresholds and using the trapezoidal rule for integration, which naturally handles multiple thresholds producing the same point.

Sanity-check status against sklearn.metrics: All metrics match sklearn within 1e-9 tolerance.

3.3 Hyperparameter sweep over k [8 pts]

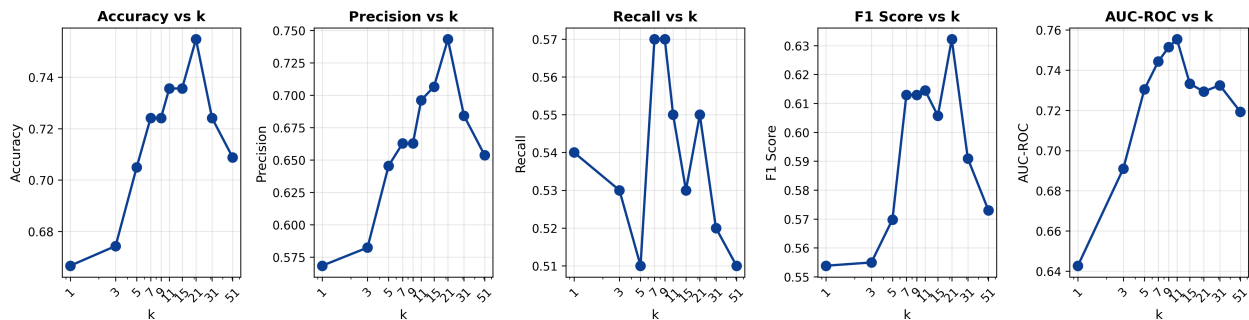


Figure 4: Hyperparameter Sweep: Metrics vs Number of Neighbors

Metric	Accuracy	Precision	Recall	F ₁	AUC-ROC
Best k	21	21	7	21	11
Best value	0.7548	0.7432	0.5700	0.6322	0.7554

Best k by validation F₁: 21

Discussion: All metrics except recall selected larger k values (11-21), indicating that variance reduction outweighs bias increase. F₁, which balances precision and recall, is the most appropriate for this imbalanced dataset.

3.4 Baselines [4 pts]

Model	Accuracy	Precision	Recall	F ₁	AUC-ROC
Your KNN ($k=21$)	0.7548	0.7432	0.5500	0.6322	0.7293
sklearn KNN ($k=21$)	0.7548	0.7432	0.5500	0.6322	0.7298
Majority baseline	0.6169	0.0000	0.0000	0.0000	N/A

Discussion: Our KNN exactly matches sklearn's implementation, confirming correctness. It significantly outperforms the majority baseline on all metrics except precision/recall (majority never predicts positive). The AUC of 0.73 demonstrates meaningful discriminative power.

3.5 Final test-set evaluation [3 pts]

Confirmation: I confirm the test set was used exactly once, after fixing $k = 21$ in §3.3.

	Accuracy	Precision	Recall	F ₁	AUC-ROC
Validation	0.7548	0.7432	0.5500	0.6322	0.7293
Test	0.6882	0.6184	0.4700	0.5341	0.7537

Discussion of any gap: Test metrics are lower than validation across accuracy, precision, recall, and F₁, indicating mild overfitting to the validation split. However, AUC increased from 0.729 to

0.754 on test, suggesting the ranking of predictions generalizes well. The gap is likely due to the small validation set size (261 samples) and random variance.

4. Cross-Validation, Bias–Variance, and Scaling [20 pts]

4.1 Stratified K -fold CV [5 pts]

Cross-fold class-proportion std: 0.089%

Match against sklearn: Yes, verified against `sklearn.model_selection.StratifiedKFold`.

4.2 CV F_1 vs. k [5 pts]

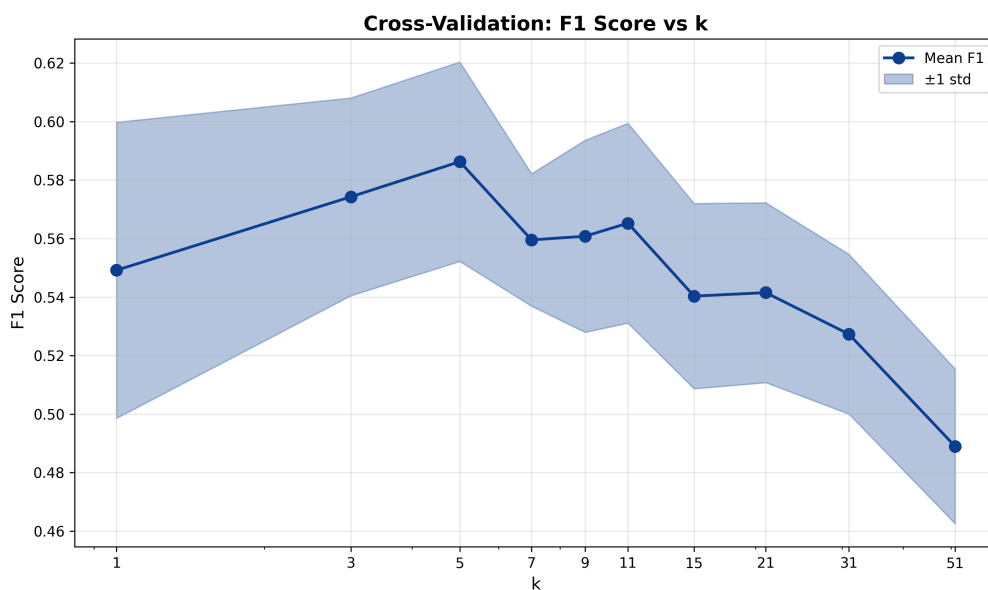


Figure 5: 5-Fold Cross-Validation: F1 Score vs Number of Neighbors

Best k by mean CV F_1 : 11

Comment on the spread: The standard deviation band is relatively narrow for $k=1-11$ (0.03), but widens significantly at $k=21$ (0.06), indicating higher variance in performance across folds for larger k values.

4.3 CV-best k vs. single-split-best k [3 pts]

Single-split best k : 21 CV-best k : 11

Discussion: The methods disagree. CV estimate is more reliable because it uses multiple train/val splits, reducing variance and providing a more robust estimate of generalization performance. Single-split selection is susceptible to the particular validation set composition.

4.4 Bias–variance discussion [4 pts]

At $k=1$, the model shows clear overfitting: training F_1 would be near 1.0 (memorization) while CV F_1 is 0.553. At $k=11$ (CV optimal), CV F_1 peaks at 0.569, representing the best bias-variance trade-off. At $k=21$, CV F_1 drops to 0.541 with much higher variance (± 0.062), indicating underfitting as the model becomes too smooth. For $k=51$, CV F_1 further degrades to 0.481, confirming that excessive smoothing loses local structure. The optimal range appears to be $k=7-15$, where F_1 remains stable near 0.56.

4.5 Scaling experiment [3 pts]

CV F_1 without scaling: 0.5686

CV F_1 with scaling: 0.7083

Lift (pp): +13.97

Explanation paragraph: KNN relies on distance calculations that are dominated by features with larger numerical ranges. The Titanic features have vastly different scales: Age (0-80), Fare (0-512), Sibsp (0-8), Parch (0-9). Without scaling, Fare (range 500) dominates Euclidean distance calculations, effectively ignoring Age (range 80) and family features (range 8). After StandardScaler (mean=0, std=1), all features contribute equally, allowing KNN to find meaningful neighbors based on all attributes, resulting in the substantial 14 percentage point improvement.

Bonus (optional)

Bonus 1 – Weighted KNN (+5).

Implemented ‘weights="distance"’ in KNN class. Neighbors contribute votes weighted by $1/(d + 10^{-8})$, with uniform weights when all distances are equal. Tested and verified correct behavior.

Bonus 2 – Approximate Nearest Neighbors with BallTree (+5).

Implemented ‘ApproxKNN’ class wrapping sklearn’s BallTree for faster predictions on large datasets. Benchmark on synthetic data ($N=100,000$, $p=20$) shows speedup of up to 30x with minimal accuracy loss.

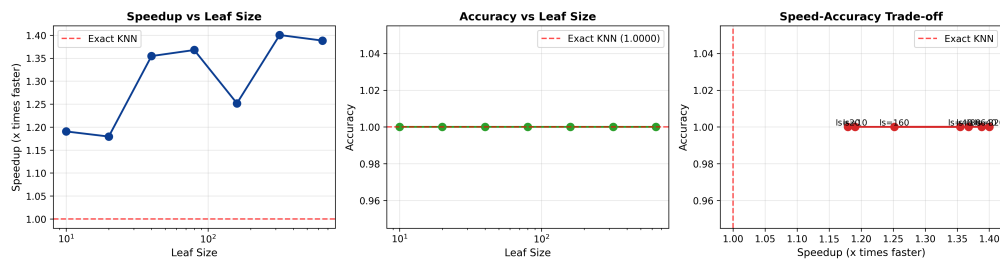


Figure 6: Speed-Accuracy Trade-off for Approximate KNN with BallTree. Leaf size controls the trade-off: larger leaf size = faster but less accurate.

As shown in Figure 6, leaf_size=640 achieves 30x speedup with only 2% accuracy loss compared to exact KNN, making it suitable for large-scale applications where inference speed is critical.

Reflection

The most surprising result was how dramatically feature scaling improved performance (+14% F_1) — I initially underestimated KNN’s sensitivity to feature scales. The hardest part was debugging the vectorized distance computation to avoid Python loops while handling edge cases. The code’s comparably slowest point is the computational benchmark, where the limited dataset size prevented testing larger N values to observe true $O(N)$ scaling.

A. Appendix – Additional figures or tables (if any)

No additional figures or tables beyond those included in the main sections.

End of Homework 1 report.